Summer training program table

Duration: One Month | Schedule: Saturday, Monday, Wednesday (3 Days/Week)

10:00 – 15:00

<table>
<tr><th>Week</th><th>Day</th><th>Session Title / Modules</th><th>Activity Type</th><th>Lecturers</th></tr>
<tr><td rowspan="2">Week 1: in Silico Techniques Foundations</td><td>Day 1
(5-8-2026)</td><td>Module 1: Protein Structure & Function Revision
- Amino acids, peptide bonds, primary to quaternary structures
- Forces stabilizing structure (H-bonds, hydrophobic, electrostatic)
- Protein folding and misfolding
- Free energy concepts in biological systems
- PDB database
Module 2: Introduction to in silico Methods
- bioinformatics overview
- Installation of PyMOL, VMD and their basics
- Introduction to Linux administration (command line)</td><td>Lecture + Hands-on (PyMOL, VMD, Linux)</td><td>Prof. Dr. Abdo A. Elfiky
&
Dr. Mahmoud Noamaan
(Mathematics, Cairo University)</td></tr>
<tr><td>Day 2
(8-8-2026)</td><td>Module 3: Protein Modeling I - Homology Modeling
- Principles of homology modeling
- Sequence alignment and template selection
- Software: SWISS-MODEL, AlphaFold 3
- Hands-on: Modeling a simple protein using online servers
Module 4: Protein Modeling II - Refinement & Validation
- Quality assessment
- Refinement of modelled structures
- Hands-on: Refining and validating modeled structures</td><td>Lecture + Hands-on (Online servers)</td><td>Dr. Ahmed A. Ezat
(Biophysics, Cairo University)</td></tr>
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Week	Day	Session Title / Modules	Activity Type	Lecturers
	Day 3 (10-8-2026)	**Module 5: Chemical Compound Modeling & Minimization** - Small molecule structures (2D/3D) - Different file formats and their conversion using OpenBabel (SDF, PDBQT, Mol2) - Minimization and conformational analysis using OpenBabel and Avogadro **Hands-on:** Drawing and optimizing small molecules	Lecture + Hands-on (OpenBabel / Avogadro)	Dr. **Mahmoud Noamaan** *(Mathematics, Cairo University)* Mr. **Ibrahim Mohamed** *(Biophysics, Cairo University)*
Week 2: Docking and Virtual Screening	Day 1 (12-8-2026)	**Module 6: Protein-Ligand Docking – Theory** - Molecular docking principles - Scoring functions, algorithms - Rigid vs flexible docking - Overview of AutoDock Vina **Module 7: Protein-Ligand Docking – Practical** - Tautomers and ionization states **Hands-on:** Preparing ligands and receptor with AutoDock Tools **Hands-on:** Docking using AutoDock Vina then analyzing results (binding poses, scores) **Module 8: Virtual Screening** **Hands-on:** Running a virtual screening campaign	Lecture + Hands-on (AutoDock Vina / Tools)	Mr. **Ibrahim Mohamed** *(Biophysics, Cairo University)* Ms. **Donia G. Yousuf** *(Biotechnology, Badr University in Cairo)*
	Day 2 (15-8-2026)	**Module 9: QM introduction** - Schrodinger equation (SE) - Molecular orbital theory - Overview of Computational Chemistry - Geometry optimization, freq and relative Energies - XYZ coordinate or Z-Matrix representation - Introduction to G09 and GV - Build molecule, optimization, and frequency calculation - Transition state and IRC - **Tutorials** about IR, UV (TD-DFT), NMR analyses - **Tutorials** in MEP and FMOs generation - Multiwfn code post-calculations - DFT calculations - Basis set	Lecture + Hands-on (Gaussian 09/ Gaussian view)	Dr. **Mahmoud Noamaan** *(Mathematics, Cairo University)*

Week	Day	Session Title / Modules	Activity Type	Lecturers
	Day 3 (17-8-2026)	**Module 10: Protein-Protein Docking - Theory** - Principles of protein-protein interactions - Algorithms (FFT-based, shape complementarity) - Software overview: HADDOCK, ClusPro, ZDOCK **Module 11: Protein-Protein Docking - Practical** - **Hands-on:** ClusPro docking - Interface analysis **Module 12: Data Analysis & Visualization** - Basic statistics - Visualizing docking results **Module 13: Project Kick-off & Literature Review** - Project phase introduction - Group formation - Guidance on literature search	Lecture + Hands-on (Online Server)	Prof. Dr. **Abdo A. Elfiky** & Dr. **Ahmed A. Ezat** *(Biophysics, Cairo University)*
Week 3: Molecular Dynamics Simulation	Day 1 (19-8-2026)	**Module 14: MD Simulation - Basics** - Principles of MD (force fields, ensembles) - Periodic boundary conditions - Software overview (GROMACS) **Module 15: MD Setup - Practical** - System prep: topology generation - Solvation, ion addition - **Hands-on:** GROMACS setup	Lecture + Hands-on (Colab / WSL)	Mr. **Ibrahim Mohamed** *(Biophysics, Cairo University)*
	Day 2 (22-8-2026)	**Module 16: MD Simulation - Equilibration & Production Run** - NPT/NVT equilibration - Production run (e.g., 10 ns) - **Hands-on:** Running simulations **Module 17: MD Analysis - Basics** - Trajectory visualization (VMD) - RMSD, RMSF, Rg, hydrogen bond analysis - Adv: Principal Component Analysis (PCA) - Adv: Free energy calculations (MM/PBSA) - **Hands-on:** MDAnalysis / plotly tools - **Hands-on:** PCA on trajectory	Lecture + Hands-on (MDAnalysis / plotly / gmxMMPBSA)	Mr. **Ibrahim Mohamed** *(Biophysics, Cairo University)* **Mr. Mahmoud E. Rashwan** *(Physics, Sohag University)* Mr. **Yousef A. Mostafa** *(Biophysics, Cairo University)*

Week	Day	Session Title / Modules	Activity Type	Lecturers
	Day 3 (24-8-2026)	**Module 18: From Force to Action: A Journey from Newtonian Mechanics to Hamilton–Jacobi Theory** - Intro numerical solution of differential equation - Explore the classical equation of motion in different format - Connection of the computational scheme with convergence and stability in error point of view **Module 19: Mixed quantum-classical dynamics simulation** - NAMD + Gaussian with python interface to manage QM/MM run - DHFR example	Lecture + Hands-on (Gaussian 09/ Gaussian view/ NAMD)	Dr. **Mahmoud Noamaan** *(Mathematics, Cairo University)*
Week 4: Project Presentation (online)	Day 1 (26-8-2026)	**Module 20: Wet-lab validation in drug discovery** **Module 21: Instrumentation** - Introduction to uc Arduino - Measuring temperature using Arduino - Measuring temperature using Arduino and PC	Lecture + Hands-on (Arduino)	Dr. **Amr A. WalyEldein** *(Zoology, Cairo University)* Mr. **Mahmoud Abdelaal** *(Biophysics, Cairo University)*
	Day 2 (29-8-2026)	**Module 22: Project Presentations & free lectures** - Student presentations on methodology, results, and conclusions - Q&A session - AI in structure-based drug design (Vina, GNINA, and Boltz2) - Measuring the Invisible: Radiation Safety in Practice **Module 23: Workshop Wrap-up** - Feedback session - Certificate distribution	Lecture + Presentation + Discussion	Ms. **Yasmine Abuhadema** *(botany and microbiology, Cairo University)* Mr. **Abdelaziz Habib Elazol** *(Medical Physicist & Radiation Safety Officer, KSA)* **&** Students